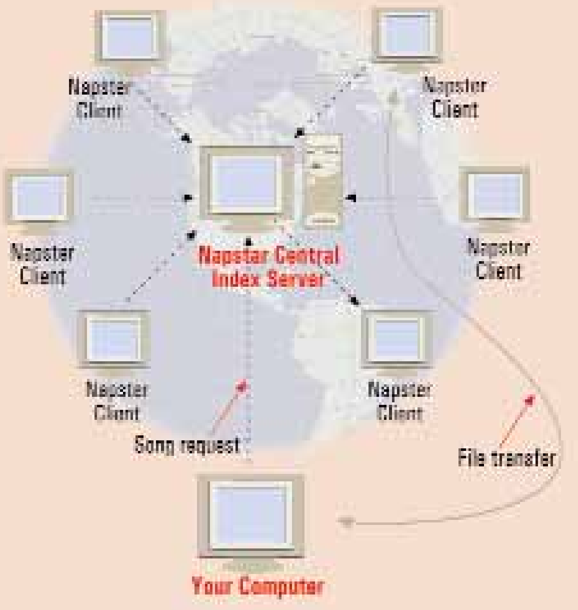
Infographics: Jaya Shetty

Napster: How it Works



Site A asks site B for data matching 'ZIP'. After passing back anything that might be of interest, site B passes the request on to its colleague (same as reference or bookmark) at site C and keeps a record that site A has made the request. If site C has something matching the request, it gives the information to site B, which remembers that it is meant for site A and passes it through to that site.

clients that could have this information. If a match is located, then the software creates a connection that allows you to download the files directly.

Gnutella is actually much more versatile than Napster because it allows users to share any type of file, not just MP3s. Also, the absence of a central index server means that the clients are almost impossible to locate. This reduces the vulnerability of the network to different types of network attacks.

However, the distributed search mechanism used by Gnutella is somewhat inefficient. Because it can only detect clients that respond to a request for information, it may not find matches that lie beyond its 'horizon', or in other words, are 'out of sight' of the software.

Freenet: an alternate Internet

Freenet, another P2P system similar to Gnutella, is working towards creating an entire alternate Internet, which tries to get around the regular structure of the Web, which involves the use of servers to exchange information. Think of it as a group of computers (individual users) connected through the Freenet software, which identifies others on the network.

When searching for specific information on Freenet, the Freenet software sends out the request in the form of a search string via the Internet to all other Freenet clients. These clients then try and match this string with the ones stored on them. If a client finds a match, it sends a response to the computer from where the request originated. This response not only contains the requested data but also the address of the node that sent the response known as 'reference'.

In case a node doesn't have the requested data but knows the reference of one that does (just like a bookmark), it passes on the request to the 'bookmarked' client. This process goes on and on till the user who made the request initially stops the search.

Spreading the Rumor: McAfee's VirusScan ASaP service

In addition to file sharing and collaboration, P2P technology is also being used to develop software that helps in improving the efficiency and reliability of software used by enterprises to protect their networks.

Rumor is a technology used in McAfee's VirusScan ASaP service, (formerly called myCIO) that allows PCs on a network to become servers and provide other clients with updated virus signature files. The first user to log on downloads an anti-virus update from the server via the Internet. Then, this user plays the role of the server for others on the network. Every time a patch is released or a virus definition is updated, the system retrieves the latest file and automatically serves it to any client that needs it.

Rumor uses a technology called token-based authentication that ensures that only legitimate anti-virus update files can be shared. This helps eliminate the possibility of rogue files spreading over the network. Another advantage that Rumor has over a system such as Napster is that it detects an incomplete download (due to a broken connection) and allows the process to be resumed later. This eliminates the likelihood of incomplete files being distributed through the network.

Another interesting feature is the application-independence of the technology. This will allow the system to be extended to deliver other file updates to users, including all kinds of security software such as firewalls and other intrusion detection software.

There are more…

The list of new P2P-based applications goes on. Groove, a corporate example, is a networking

The Humane Side

Intel and United Devices have set up a 'virtual supercomputer' to aid researchers optimising a drug for curing leukemia.

The virtual supercomputer, which is quite similar to SETi which is being used to search for extraterrestrial intelligence, will use a concept called distributes computing that allows utilisation of unused processing power and hard disk space of millions of computers from across the Internet. Each computer downloads a small application that analyses the cancer-fighting properties of 250 million molecules. When finished, it sends processed data back to researchers, for further analysis.

Depending on how many people choose to participate, the time required to develop new drugs could be cut down to as little as five years from the current estimate of 12 years.

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